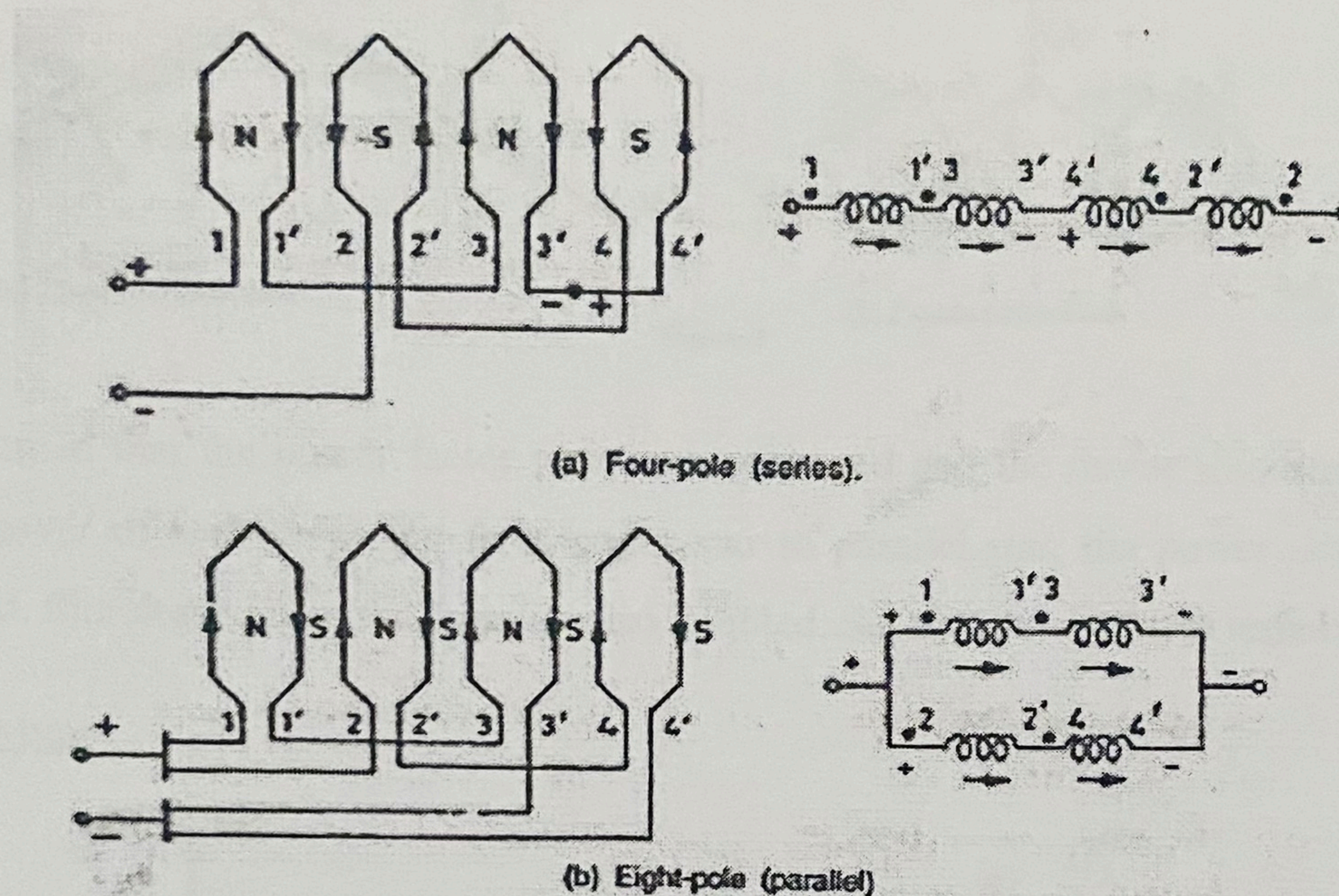




Figure 1 (a) shows schematically only four coils of one phase of the connected in series, along with the direction of current in them, producing four poles in the stator. If the current in coils 2 and 4 is reversed and the connection is changed to parallel with two coils (1 and 3, and 2 and 4) connected in series for each path, eight poles are formed in the stator (Figure 1 (b)). It may be noted that the direction of current in coils 1 and 3 remains the same. Only one type of connection is shown.



Constant torque and constant power operations

The choice of winding arrangements of 3 phase, two speed motors depends on the operating characteristics required at the two speeds. For constant torque operation, the change of stator winding is made from series-star to parallel-star, while for constant power operation the change is made from series-delta to parallel-delta,

a) Constant torque type

If it is desired to have constant torque at both the speeds, the arrangement is as shown in figure 2. For low speed, the voltage is applied to terminals $U_1-V_1-W_1$ while $U_2-V_2-W_2$ are left open and for high speed operation of the motor, the voltage is applied to terminals $U_2-V_2-W_2$ and the terminals $U_1-V_1-W_1$ are shorted. The flux densities for the two speeds are approximately equal; the motor can be considered to have a constant torque. The output of the motor is approximately proportional to the speed.

Let V = line voltage, I = Maximum current that the winding can carry. Then, the power drawn from the supply is given by,

$$\text{For series - star connection, } P_Y = \sqrt{3}VI \cos \phi_Y$$

$$\text{For parallel - star connection, } P_Y = 2\sqrt{3}VI \cos \phi_{Y'}$$